Abstract maps and their coordinate matrices

A linear transformation is a map between vector spaces

$$T: V \to W,$$

that preserves addition and scalar multiplication. The map itself does not require a basis. Choosing domain and codomain bases gives its matrix representation.

Rotation matrix

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

is the representation of rotation in standard coordinates. Changing bases changes the entries without changing the underlying rotation.

Kernel, image, rank, injectivity, and surjectivity are intrinsic to the map. Matrix entries depend on coordinates. For determinants, invariance holds under a common basis change for an endomorphism; independent domain and codomain changes alter the coordinate volume factors.

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Linearity

A map $T : V \to W$ is linear if, for every u, v and scalars α, β,

$$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v).$$

Hence $T(0) = 0$, and for every finite linear combination,

$$T\left(\sum_i c_i v_i\right) = \sum_i c_i T(v_i).$$

Knowing the images of a basis determines the map. The condition $T(0) = 0$ is necessary but does not alone prove linearity.

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Kernel, image, and rank-nullity

Define kernel and image by

$$N(T) = \{v \in V : T(v) = 0\},$$
$$C(T) = \{T(v) : v \in V\}.$$

They are subspaces of V and W. For finite-dimensional V, rank-nullity states

$$\dim V = \dim N(T) + \dim C(T).$$

To prove it, take a kernel basis

$$v_1, \ldots, v_k,$$

and extend it to a basis of V.

$$v_1, \ldots, v_k, u_1, \ldots, u_r.$$

The vectors $T(u_1), \ldots, T(u_r)$ span the image by applying T to the full basis expansion. If $\sum_i c_i T(u_i) = 0$, then $\sum_i c_i u_i$ lies in the kernel. Independence of the extended basis forces every $c_i = 0$. Thus these image vectors form a basis.

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Injectivity, surjectivity, and dimension

A linear map is injective exactly when

$$N(T) = \{0\}.$$

If the kernel is trivial, $T(u) = T(v)$ implies $T(u - v) = 0$, hence $u = v$. Conversely, injectivity and $T(v) = T(0)$ imply $v = 0$.

Surjectivity means

$$C(T) = W.$$

For a finite-dimensional map $T\colon V \to V$, the following are equivalent.

- T invertible;
- The kernel contains only zero.
- The rank equals $\dim V$.
- coordinate matrix invertible.

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Injectivity, surjectivity, and dimension (continued)

With unequal dimensions, injectivity and surjectivity differ. A map $\mathbb{R}^2 \to \mathbb{R}^3$ can be injective, but its image has dimension at most 2 and cannot fill the codomain.

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Examples of linear maps

- Rotation $T(x, y) = (-y, x)$ has zero kernel and the whole plane as image.
- Projection $P(x, y) = (x, 0)$ has the y-axis as kernel and the x-axis as image.
- On real- or complex-coefficient $\mathcal{P}_k$, differentiation has constant kernel; its image is $\mathcal{P}_{k-1}$ for $k \geq 1$, and $\{0\}$ for $k = 0$.
- $I(p)(t) = \int_0^t p(s)\, ds$ has zero kernel and as image the polynomials of degree at most $k + 1$ with zero constant term.
- Scaling $S(x) = cx$ is invertible for $c \neq 0$; for $c = 0$ its image is zero.

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Linear and affine maps

Affine map:

$$T(x) = Ax + b.$$

An affine map preserves affine combinations, $T((1-t)x + ty) = (1-t)T(x) + tT(y)$. A line can collapse to a point if its direction lies in the kernel of A; otherwise its along-line ratios are preserved. For $b \neq 0$,

$$T(0) = b \neq 0,$$

so the map is not linear on the original vector space.

A straight-line graph need not describe a linear map. For example,

$$f(x) = 2x + 1$$

fails $f(0) = 0$ and the scalar multiplication rule.

The map $T(x) = x^2$ is also nonlinear, failing addition and scalar multiplication. Testing $T(0) = 0$ can reject a map immediately, but passing that test alone does not establish linearity.

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Matrix columns from basis images

Given ordered input basis $(v_1, \ldots, v_n)$ and output basis $(w_1, \ldots, w_m)$, expand each $T(v_j) = \sum_i M_{ij} w_i$.
Column j is the output-coordinate vector of the image of input basis vector j:

$$M = \begin{bmatrix} [T(v_1)]_{\text{out}} & \cdots & [T(v_n)]_{\text{out}} \end{bmatrix}.$$

For $x = \sum_j c_j v_j$, linearity gives

$$[T(x)]_{\text{out}} = M[x]_{\text{in}}.$$

M has m rows and n columns; coordinate order must match the ordered bases.

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A complete standard-matrix example

Let

$$T(x, y) = (x + y, x - y, 2x).$$

The images of the standard basis vectors are

$$T(e_1) = (1, 1, 2)^T, \qquad T(e_2) = (1, -1, 0)^T.$$

so

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 0 \end{bmatrix}.$$

This matrix is 3×2. Its kernel satisfies

$$x + y = 0, \quad x - y = 0, \quad 2x = 0$$

forcing $x = y = 0$, so T is injective. Its image is a rank-2 plane in $\mathbb{R}^3$, not the entire codomain.

Now let $S \colon \mathbb{R}^3 \to \mathbb{R}^2$,

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A complete standard-matrix example (continued)

$$S(x,y,z) = (x+z, y-2z), \qquad N = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \end{bmatrix}.$$

Composition means $(S \circ T)(v) = S(T(v))$. When the output basis of T matches the input basis of S,

$$[S \circ T]_{\text{std}} = NA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ -3 & -1 \end{bmatrix}.$$

For $v = (1,2)^T$, $T(v) = (3,-1,2)^T$, so

$$S(T(v)) = (5,-5)^T = (NA)v.$$

The rightmost factor A acts first, followed by N. Insert a coordinate change if the intermediate bases differ.

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The change-of-basis formula

Let $V = \mathbb{R}^n$, $W = \mathbb{R}^m$, and A be the standard-coordinate matrix. The columns of the invertible square matrices $B_{\text{in}}, B_{\text{out}}$ are the new basis vectors in the respective standard coordinates.

$$[x]_{\text{std}} = B_{\text{in}}[x]_{B_{\text{in}}},$$
$$[y]_{B_{\text{out}}} = B_{\text{out}}^{-1} y_{\text{std}}.$$

Apply these operations in order.

1. Convert input basis coordinates to standard coordinates with B_{in}.
2. Apply A.
3. Convert the output to its new basis coordinates with B_{out}^{-1}.

Consequently,

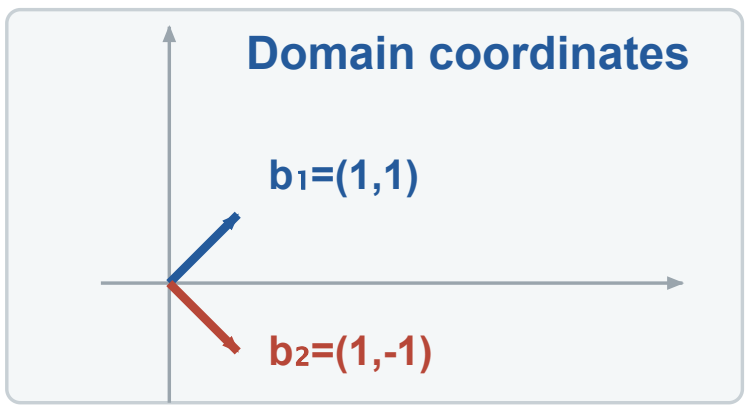
$$[T(x)]_{B_{\text{out}}} = B_{\text{out}}^{-1} A B_{\text{in}} [x]_{B_{\text{in}}},$$
$$M = B_{\text{out}}^{-1} A B_{\text{in}}.$$

The domain and codomain bases can have different dimensions.

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The change-of-basis formula (continued)

Basis conversion: Bin at input, Bout⁻¹ at output



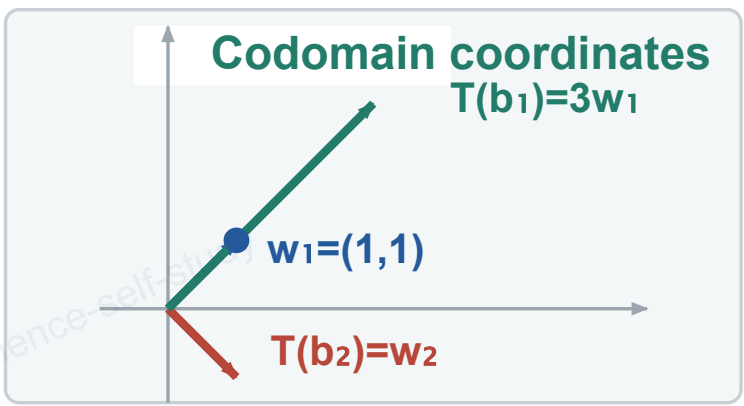
A=[[2,1],[1,2]]

Bin=[[1,1],[1,-1]]

Bout=Bin

M=[[3,0],[0,1]]

M=Bout⁻¹ A Bin



For a two-dimensional example, take

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad B_{\text{in}} = B_{\text{out}} = B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Since $Ab_1 = (3,3)^T = 3b_1$、$Ab_2 = (1,-1)^T = b_2$，

The change-of-basis formula (continued)

$$M = B^{-1}AB = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

for $x = (2, 0)^T$, the input coordinates are $B^{-1}x = (1, 1)^T$. Applying M gives $(3, 1)^T$, and converting back with B gives

$$B(3, 1)^T = (4, 2)^T = A(2, 0)^T.$$

The input conversion uses B_{in}, while the output conversion uses B_{out}^{-1}. These directions must not be reversed.

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Different input and output dimensions still permit basis changes

Use the standard matrix $A \in \mathbb{R}^{3\times 2}$ of $T(x, y) = (x + y, x - y, 2x)$. Put the new basis vectors into columns:

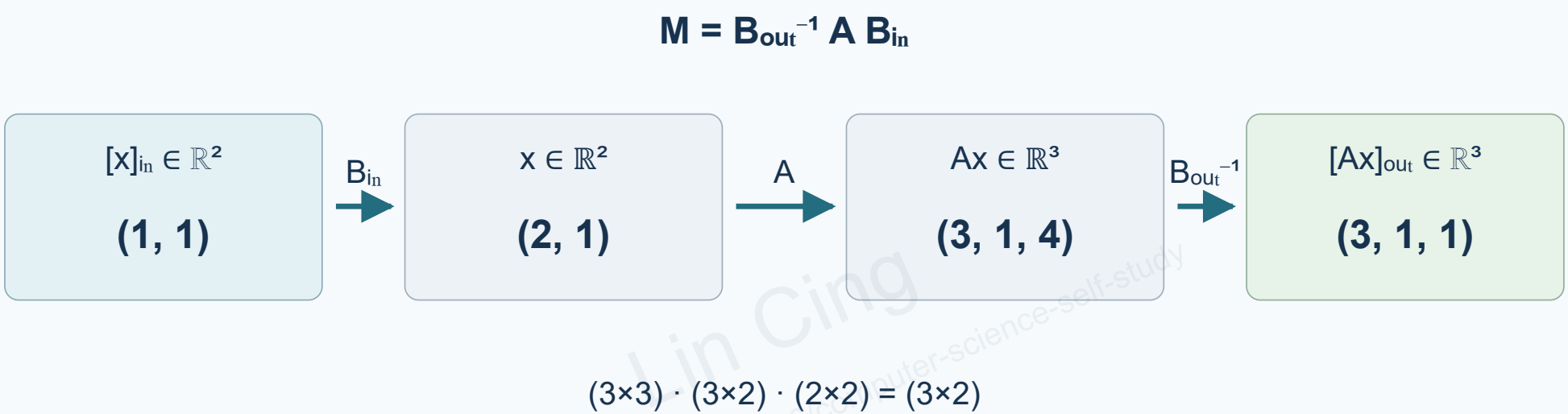
$$B_{\text{in}} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad B_{\text{out}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}.$$

These are invertible 2×2 and 3×3 basis matrices. Multiplying gives $AB_{\text{in}} = \begin{bmatrix} 1 & 2 \\ 1 & 0 \\ 2 & 2 \end{bmatrix}$, and

$$B_{\text{out}}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}, \quad M = B_{\text{out}}^{-1} A B_{\text{in}} = \begin{bmatrix} 1 & 2 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}.$$

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Check all four coordinate vectors on the same input



Start with $[x]_{\text{in}} = (1, 1)^T$. From left to right, the standard input is $x = (2, 1)^T$, the standard output is $Ax = (3, 1, 4)^T$, and $[Ax]_{\text{out}} = (3, 1, 1)^T$. Direct multiplication by M gives the same result; both dimensions and values agree.

Similarity on one space

When $V = W$ and the same new basis B is used on both sides,

$$[T]_B = B^{-1}AB.$$

If $Av = \lambda v$ and $y = B^{-1}v$,

$$B^{-1}AB\,y = B^{-1}Av = \lambda y.$$

so the eigenvalues agree. More fully, $\lambda I - B^{-1}AB = B^{-1}(\lambda I - A)B$, proving equality of characteristic polynomials. Trace and determinant are also invariant.

$$\operatorname{tr}(B^{-1}AB) = \operatorname{tr} A,$$
$$\det(B^{-1}AB) = \det A.$$

Independent input and output bases need not give similarity. The resulting matrix can be rectangular, and even square matrices related by independent left and right changes need not be similar.

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Even square coordinate matrices need not be similar

Let $T = I_2$, with input basis matrix $B_{\text{in}} = I_2$ and output basis matrix $B_{\text{out}} = 2I_2$. The map still returns the same vector, but its new coordinate matrix is

$$M = B_{\text{out}}^{-1} I_2 B_{\text{in}} = \tfrac{1}{2} I_2.$$

The eigenvalues of M are $1/2, 1/2$, and its determinant is $1/4$; for I_2, both eigenvalues and the determinant are 1. The matrices are not similar.

The similarity formula uses the **same** new basis for input and output in one space. Independently changing the two bases does not imply similarity, even when the matrices are square.

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An eigenbasis example

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

Eigenbasis:

$$B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

$$B^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Therefore

$$B^{-1}AB = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}.$$

In this basis, the same map applies independent scalings. An ill-conditioned eigenbasis can amplify coordinate errors, so exact diagonalization does not itself guarantee a good numerical representation.

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Differentiation in two bases

$D : \mathcal{P}_2 \to \mathcal{P}_1$, input monomial basis

$$(1, t, t^2),$$

output basis $(1, t)$.

$$D(1) = 0, \qquad D(t) = 1, \qquad D(t^2) = 2t.$$

Hence

$$M = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

If the input basis becomes $1, s, s^2$, where $s = t - 1$, then

$$D(s^2) = 2s = -2 + 2t,$$

the new coordinate matrix is

$$\begin{bmatrix} 0 & 1 & -2 \\ 0 & 0 & 2 \end{bmatrix}.$$

The representation changes, while differentiation itself does not.

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A map on polynomials: start with the basis images

Let $T\colon \mathcal{P}_2 \to \mathcal{P}_2, T(p) = xp' + p''$, with ordered bases $\beta = (1, x, x^2)$, $\gamma = (1, 1+x, 1+x^2)$.

Since $T(1) = 0$, $T(x) = x$, and $T(x^2) = 2 + 2x^2$, the matrix in β and the change-of-coordinates matrix are

$$A = [T]_\beta = \begin{bmatrix} 0 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \quad P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The columns of P are the β-coordinates of the γ vectors, so $[p]_\beta = P[p]_\gamma$; this direction matters.

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Compute the same coordinate matrix in two ways

$P^{-1} = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, giving

$$[T]_\gamma = P^{-1}AP = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Check columns directly: $T(1) = 0$, $T(1+x) = x = -1 \cdot 1 + 1(1+x)$, and $T(1+x^2) = 2(1+x^2)$.

For $[p]_\gamma = (2, -1, 3)^T$, $[p]_\beta = P[p]_\gamma = (4, -1, 3)^T$, so $p = 4 - x + 3x^2$. Then $T(p) = 6 - x + 6x^2$, with γ-coordinates $(1, -1, 6)^T$, exactly as obtained from $[T]_\gamma[p]_\gamma$.

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Invariant subspaces

A subspace $U \subseteq V$ is invariant if

$$T(U) \subseteq U.$$

An eigenvector line is a one-dimensional nonzero invariant subspace. If

$$V = U \oplus W$$

and both summands are invariant, the matrix is block diagonal in a basis adapted to the decomposition.

If only U is known invariant, begin a basis with a basis of U. The matrix has block upper triangular form

$$\begin{bmatrix} A_1 & B \\ 0 & A_2 \end{bmatrix}.$$

Invariant subspaces reduce a large map to smaller blocks. Both Schur and Jordan forms use this structure.

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Eigenbases and diagonalization

If $b_1, \ldots, b_n$ form an eigenbasis, let

$$B = [b_1 \ \cdots \ b_n],$$

Then

$$B^{-1}AB = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

The eigenbasis separates coupled equations into scalar modes, simplifying powers, exponentials, and recurrences.

For a real symmetric matrix, choose an orthonormal eigenbasis Q, giving

$$Q^{-1} = Q^T.$$

Inversion becomes transposition and the basis condition number is 1. A nonsymmetric eigenbasis can be nearly dependent despite existing exactly.

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Jordan form and generalized eigenvectors

Every complex square matrix has a Jordan form

$$J = B^{-1}AB.$$

whose blocks have the form

$$J_k(\lambda) = \begin{bmatrix} \lambda & 1 & & 0 \\ 0 & \lambda & \ddots & \\ & \ddots & \ddots & 1 \\ 0 & & 0 & \lambda \end{bmatrix}.$$

The diagonal contains eigenvalues; superdiagonal ones encode generalized-eigenvector chains. A nonzero vector w satisfying

$$(A - \lambda I)^m w = 0$$

for some $m \geq 1$ is a generalized eigenvector. A Jordan chain satisfies $(A - \lambda I)w_1 = 0$, $(A - \lambda I)w_{j+1} = w_j$, giving the superdiagonal ones. The general Jordan existence theorem is stated without its full proof here; the exercises compute a second-order nilpotent block explicitly.

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Jordan form and generalized eigenvectors (continued)

Jordan form is a theoretical classification. Numerical algorithms usually use Schur form, SVD, or structured methods because computing Jordan structure is unstable.

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Circulant matrices and a Fourier basis

For $n = 4$, the left cyclic shift

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

maps $(x_0, x_1, x_2, x_3)^T$ to $(x_1, x_2, x_3, x_0)^T$.

Choose roots of unity

$$\lambda \in \{1, i, -1, -i\},$$
$$v_\lambda = (1, \lambda, \lambda^2, \lambda^3)^T.$$

The wraparound coordinate uses $\lambda^4 = 1$, so

$$Pv_\lambda = \lambda v_\lambda.$$

The four distinct eigenvalues give four independent eigenvectors. The Fourier matrix diagonalizes P and every circulant polynomial

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Circulant matrices and a Fourier basis (continued)

$$C = c_0 I + c_1 P + c_2 P^2 + c_3 P^3.$$

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Distinguish a finite basis from an infinite expansion

In $\mathcal{P}_n$, $(1, x, \ldots, x^n)$ is a finite basis. Monomials are also a Hamel basis of the space of polynomials, because each polynomial uses finitely many coefficients. Finite monomial combinations do not represent every continuous function, such as e^x.

Fourier exponentials form a complete orthogonal expansion system in $L^2[-\pi, \pi]$: partial sums approximate each element in the L^2 norm. This **infinite expansion** with a specified convergence sense does not make every function a finite combination and does not by itself guarantee pointwise convergence. Legendre polynomials similarly give orthogonal expansions in $L^2[-1, 1]$; a finite truncation lies in a finite-dimensional subspace.

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First identify the direction of error propagation

Let an invertible basis matrix satisfy $x = Xc$. A geometric perturbation δx changes the coordinates by

$$\delta c = X^{-1}\delta x, \qquad \|\delta c\|_2 \le \|X^{-1}\|_2 \|\delta x\|_2.$$

Conversely, a coordinate perturbation gives $\delta x = X\delta c$, controlled by $\|X\|_2$. The two directions have different amplification factors.

For $x, c \neq 0$, using $\|x\| \le \|X\|\|c\|$ yields the relative bound

$$\frac{\|\delta c\|_2}{\|c\|_2} \le \kappa_2(X) \frac{\|\delta x\|_2}{\|x\|_2}.$$

A large condition number indicates possible worst-case sensitivity; not every direction or perturbation attains the bound.

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A combined example

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

$$B^{-1}AB = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}.$$

For

$$x = (2, 0)^T,$$

eigenbasis coordinates:

$$c = B^{-1}x = (1, 1)^T.$$

$$[T^5 x]_B = \begin{bmatrix} 4^5 & 0 \\ 0 & 2^5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1024 \\ 32 \end{bmatrix}.$$

return to standard coordinates by

$$T^5 x = B \begin{bmatrix} 1024 \\ 32 \end{bmatrix} = \begin{bmatrix} 1056 \\ 992 \end{bmatrix}.$$

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A combined example (continued)

The eigenbasis separates the two modes. An ill-conditioned basis would require an additional coordinate-error analysis.

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Chapter review

- Kernel and image describe the map; its matrix depends on ordered input and output bases.
- Convert input coordinates to reference coordinates, apply the map, then convert the output.
- Similarity uses the same new basis on both sides; independent changes do not guarantee it.
- Infinite expansions require a function space and convergence sense; basis conditioning affects coordinate sensitivity.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 8.

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